Daily Logic Drill #7

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mixed Synthesis Capstone · Advanced Quantifier Analysis · Multi-Claim Evaluation · BMO1 Proof Pressure.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Synthesis of foundational logic principles under speed pressure.

1. State the contrapositive of: "If x is prime and $x > 2$, then x is odd."

2. What is the negation of: "For all real numbers x, there exists a real number y such that $x + y = 0$."

3. True or false: "If a conditional statement $P \implies Q$ is true, its converse $Q \implies P$ must also be true."

4. Which of the following is equivalent to $\neg(P \implies Q)$? (A) $\neg P \implies \neg Q$, (B) $P \wedge \neg Q$, (C) $\neg P \vee Q$.

5. What single counterexample disproves: "$n^2 + n + 17$ is prime for all non-negative integers n?"

6. If $A \implies B$ and $B \implies C$, what logical rule allows concluding $A \implies C$?

7. True or false: "An assumption that leads to $0 = 1$ in a proof by contradiction proves the original premise false."

8. Give a counterexample to: "If $x^2 > y^2$, then $x > y$."

9. What is the base case needed for an inductive proof that $3^n > 2^n + 10$ for $n \geq 3$?

10. True or false: "The negation of 'every student passed' is 'every student failed'."

Section B Manipulation Drills (≤50 s each)

Direct proofs, contrapositives, and counterexample constructions.

1. Prove by contrapositive: "If n^2 is an odd integer, then n is an odd integer."

2. Find all integer counterexamples in $-10 \leq x \leq 10$ to: "If $x^2 > 4$, then $x > 2$."

3. Prove by contradiction: "$\sqrt{3}$ is an irrational number."

4. Classify the condition "$x = 2$" as necessary, sufficient, both, or neither for "$x^2 - 5x + 6 = 0$".

5. Classify the condition "$x^2 - 5x + 6 = 0$" as necessary, sufficient, both, or neither for "$x = 2$".

6. Prove by induction: $\sum_{i=1}^{n} 2^i = 2^{n+1} - 2$ for all positive integers n.

7. Give a counterexample to: "If a, b are irrational, then a^b is irrational."

8. Prove by contradiction: "If n is an integer and $n^3 + 5$ is odd, then n is even."

9. State the negation of: "For every $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$, then $|f(x) - L| < \varepsilon$."

10. Prove by induction that $7^n - 1$ is divisible by 6 for all positive integers n.

Section C Substitution & Structure (≤90 s each)

Multi-claim evaluations and tricky competition MCQs.

1. For real numbers x, y, consider: *(after TMUA 2022 Paper 2 Q10)*
I. $\forall x\, \exists y\ (x + y = 0)$
II. $\exists y\, \forall x\ (x + y = 0)$
III. $\forall x\, \forall y\ (x + y = 0)$
Which of I, II, III are true?

A) none

B) I only

C) II only

D) III only

E) I and II only

F) I, II and III

2. Determine the complete set of real values k for which the statement: "For all real x, if $x > k$, then $x^2 > k$" is true. *(after TMUA 2022 Paper 2 Q9)*

A) $k \geq 1$ or $k \leq 0$

B) $k \geq 1$ only

C) $k \leq 0$ only

D) all real numbers k

E) no real numbers k

3. Consider positive integers a, b. Statement P: "$a + b$ is even." Statement Q: "$a^2 + b^2$ is even." Which of the following is true?

A) P is necessary but not sufficient for Q.

B) P is sufficient but not necessary for Q.

C) P is necessary and sufficient for Q.

D) P is neither necessary nor sufficient for Q.

E) $P \implies \neg Q$.

4. A student attempts to prove: "There are no integer solutions to $x^2 + y^2 = 7$."
Line 1: $x^2 \pmod 4 \in \{0, 1\}$ and $y^2 \pmod 4 \in \{0, 1\}$.
Line 2: The possible values of $(x^2 + y^2) \pmod 4$ are $0 + 0 = 0, 0 + 1 = 1, 1 + 1 = 2$.
Line 3: But $7 \equiv 3 \pmod 4$.
Line 4: Since $3 \notin \{0, 1, 2\}$, $x^2 + y^2 = 7$ has no integer solutions.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 1 is false because $x^2 \pmod 4$ can equal 2.

C) Line 2 is false because $1 + 1 = 3 \pmod 4$.

D) Line 3 is false because $7 \equiv 1 \pmod 4$.

E) Line 4 is invalid because modular arithmetic cannot be used for Diophantine equations.

5. For a real number x, let P: "$x^2 - 4 = 0$" and Q: "$x = 2$". Which of the following statements is true?

A) $P \implies Q$ is true, but $Q \implies P$ is false.

B) $Q \implies P$ is true, but $P \implies Q$ is false.

C) $P \iff Q$ is true.

D) Both $P \implies Q$ and $Q \implies P$ are false.

E) $\neg P \implies \neg Q$ is false.

6. Consider the claim: "For every positive integer n, $n^3 + 2n$ is divisible by 3."
Which proof technique is most direct?

A) Factorise $n^3 + 2n = n(n^2 + 2) = n(n^2 - 1 + 3) = n(n - 1)(n + 1) + 3n$, where $(n - 1)n(n + 1)$ is divisible by 3.

B) Prove by contradiction assuming $n^3 + 2n$ is even.

C) Use the binomial theorem on $(n + 1)^3$.

D) Disprove by counterexample at $n = 3$.

E) Prove the converse for n^2.

7. What is the negation of "All prime numbers are odd or equal to 2"?

A) All prime numbers are even and not equal to 2.

B) There exists a prime number that is even and not equal to 2.

C) There exists a prime number that is odd and equal to 2.

D) No prime numbers are odd.

E) Every prime number is composite.

8. Let S be a set of n distinct positive integers. Consider the statement: "If $n \geq 5$, then S contains two numbers whose difference is a multiple of 4." Is this statement true or false? *(after SMC 2017 Q18)*

A) True, by the Pigeonhole Principle (4 possible remainders mod 4).

B) False, counterexample $\{1, 2, 3, 4, 5\}$.

C) False, counterexample $\{1, 3, 5, 7, 9\}$.

D) True, only for $n \geq 8$.

E) False, because remainders mod 4 can be negative.

Section D Challenge (BMO1 Capstone difficulty)

Full BMO1 & TMUA competition capstone proofs.

1. Prove by contradiction: "There do not exist integers x, y such that $x^2 - 5y^2 = 3$." *(after BMO1 2015 Q1)*
(Hint: Analyze the equation modulo 5.)

2. A student claims: "If a, b, c are positive integers such that $a \mid bc$, then $a \mid b$ or $a \mid c$." *(after TMUA 2019 Paper 2 Q7)*
(a) Provide a counterexample to show the claim is false in general.
(b) Add the minimal extra condition on a and b (or a and c) that makes the claim true, and state the name of this classic number-theoretic result.

3. Prove by mathematical induction that for all positive integers n,

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n+1)! - 1.$$

4. Consider three statements about a real function $f : \mathbb{R} \to \mathbb{R}$: *(after TMUA 2023 Paper 2 Q12)*
I. If $f(x) > 0$ for all $x > 0$, then $f(1) > 0$.
II. If $f(1) > 0$, then $f(x) > 0$ for all $x > 0$.
III. If $f(1) \leq 0$, then $f(x) \leq 0$ for some $x > 0$.
Which of the statements I, II, III are logically equivalent to each other?

5. Prove by contradiction that there are infinitely many prime numbers (Euclid's Theorem). Specifically, assume there are only finitely many primes $p_1, p_2, \ldots, p_k$, consider $N = p_1 p_2 \ldots p_k + 1$, and derive a contradiction regarding the prime factors of N.

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths